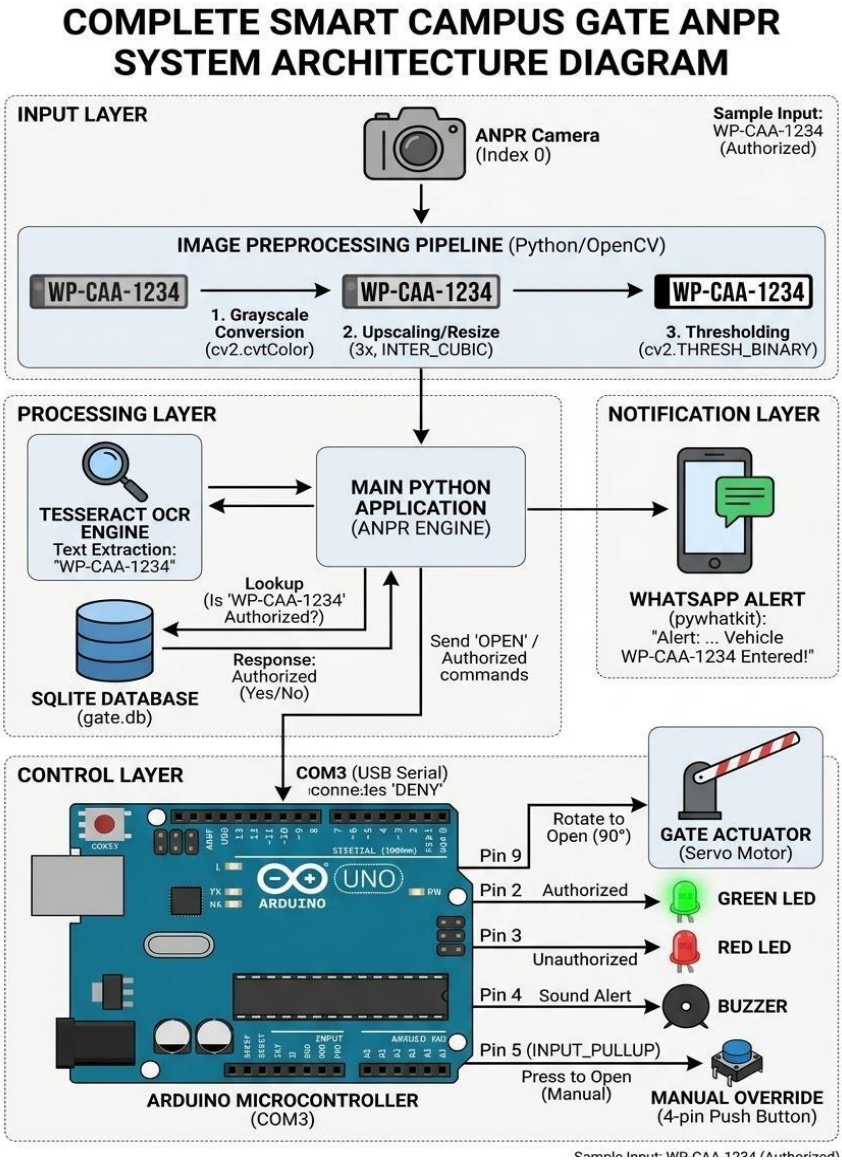
Project Name : Smart Gate Project Group

ABOUT OUR PROJECT

The Smart Campus Gate ANPR system is an advanced, automated vehicle access control solution engineered to enhance security and streamline entry management at campus perimeters. By seamlessly integrating computer vision algorithms powered by OpenCV and Tesseract OCR, the system captures and reads incoming vehicle license plates in real-time. This extracted data is instantly cross-referenced against a secure local SQLite database to verify whether the vehicle is authorized for entry. Upon successful verification, the system communicates via serial interface with an Arduino Uno microcontroller, which orchestrates the physical hardware components—triggering a servo motor to open the gate barrier, illuminating status LEDs, and sounding an audio alert. Furthermore, the system enhances administrative oversight by automatically dispatching instant WhatsApp notifications to designated mobile devices via pywhatkit, while also incorporating a manual push-button override to ensure seamless operational reliability under any circumstance.



Hardware Components (Arduino & Circuitry)

Microcontroller: Arduino Uno connected via USB Serial Communication (COM3).

Gate Actuator: Servo Motor (connected to Pin 9) to automatically open and close the campus gate barrier.

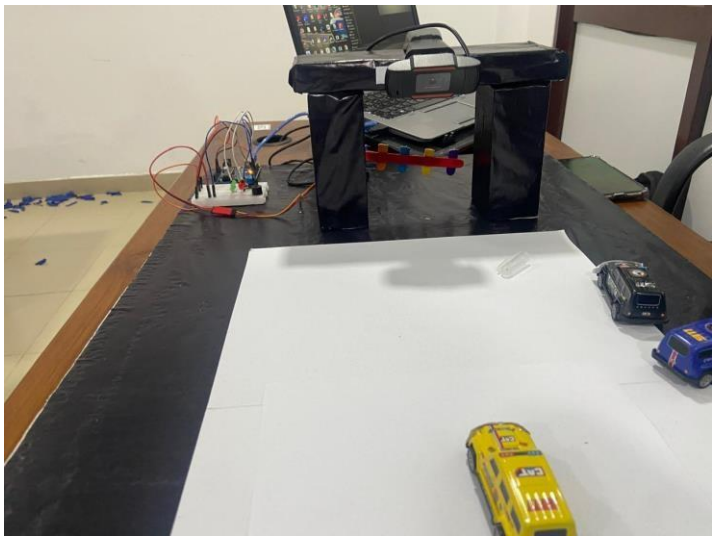
Status Indicators: Green LED (Pin 2): Indicates access granted / authorized vehicle.

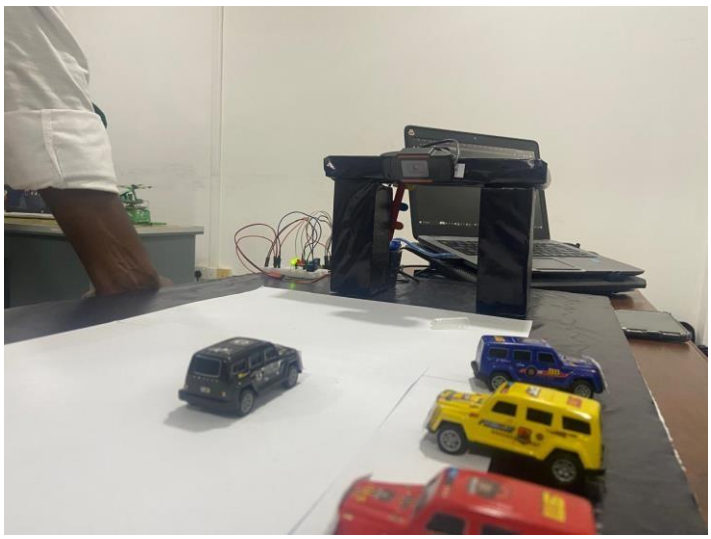
Red LED (Pin 3): Indicates access denied / unauthorized vehicle.

Audio Feedback: Buzzer (Pin 4) to sound alerts upon vehicle detection or entry.

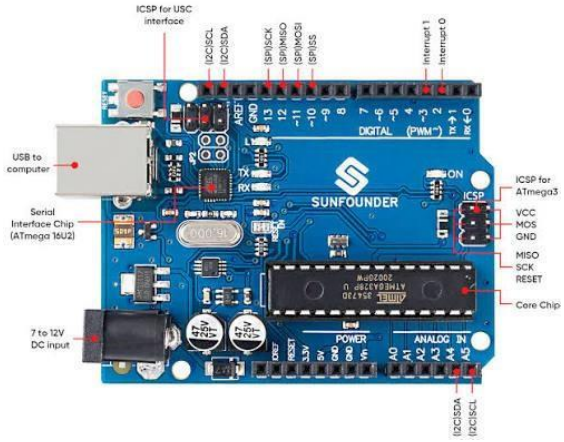
Manual Override: 4-pin Push Button (connected to Pin 5 using INPUT_PULLUP) to manually open the gate when needed, bypassing the automated OCR check.

Breadboard (Solderless Test Board): Used as the base structure to easily connect all the electronic components and jumper wires without soldering.





1. Arduino Uno Microcontroller



Role: The physical brain and central controller of the hardware circuit.

Function: It communicates with your laptop via USB Serial Communication on COM3. It listens for incoming serial commands ("OPEN" or "DENY") sent by the Python script and translates them into electrical signals to control the connected components.

2. Servo Motor (Gate Barrier Actuator)



Pin Connection: Connected to Pin 9.

rotates the servo (typically 90 degrees) to lift the gate barrier, allowing the vehicle to pass through before returning it to the closed position.

3. Status LEDs (Visual Indicators)



Green LED (Pin 2): Lights up instantly when an authorized vehicle is successfully recognized, providing a clear visual confirmation of valid access.

Red LED (Pin 3): Lights up when an unrecognized or unauthorized license plate is detected, signaling that access has been denied.

4. Buzzer (Audio Alert)



Pin Connection: Connected to Pin 4.

Function: Emits a sound alert whenever a vehicle is processed or when an entry event occurs, ensuring both visual and auditory feedback at the gate.

5. Manual Override Push Button



Pin Connection: Connected to Pin 5 using the internal pull-up configuration (INPUT_PULLUP).

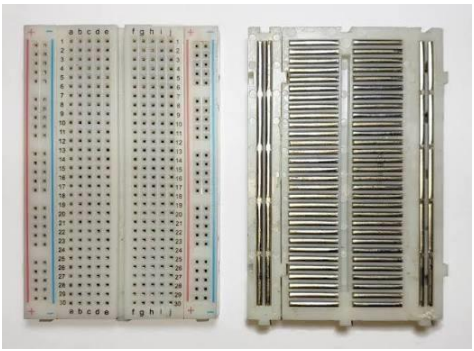
Function: Acts as a physical backup control. Pressing this button overrides the automated OCR and database check, sending a direct signal to the Arduino to open the gate manually during system maintenance, network issues, or emergencies.

6. ANPR Camera



Function: Mounted at the campus entrance to capture real-time video frames of incoming vehicles. It feeds the raw image data directly into the Python/OpenCV script so the license plate can be cropped and processed.

7. Breadboard



Core Function: The Breadboard acts as a central solderless connection hub that links all electronic components to the Arduino Uno cleanly and safely.

Power Distribution: Shares the 5V and GND power from the Arduino across its power rails

(+ / -) to supply the LEDs, Buzzer, and Push Button.

Component Organization: Holds the Green & Red LEDs (Pins 2 & 3) with resistors, the Buzzer (Pin 4), and the Manual Override Push Button (Pin 5) securely without permanent soldering.

Signal Routing: Connects the digital control pins of the Arduino to each component using jumper wires.

Testing & Modification: Allows quick wire adjustments and easy circuit troubleshooting during system testing.